

HIRAK MONDAL

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Professional Summary

Backend Engineer with **3+ years** of experience building payment systems at PayU, working across **microservices**, **Kafka**, and **AWS**. Currently own **wallet infrastructure** processing **INR 10Cr+ transactions** at a **99.6% success rate** and scaled to **2500 TPS**. Delivered measurable performance gains, including **87% faster DB queries**, **23% faster APIs** through **Redis caching**, and an **80–90% reduction in infrastructure costs**. Experienced in **Java**, **Python**, **PHP**, **encryption & security**, and **observability** tooling like Grafana and OpenSearch.

Experience

PayU Payments Private Limited — India's leading online payment platform

Bengaluru, India

Software Engineer

Apr 2025 – Present

- Took **end-to-end ownership** of the critical **StoreCard** project, optimizing **DB queries** to cut **query execution time by 87%** and introducing a **Redis caching layer** that made critical APIs **23% faster**.
- Removed **external JAR dependencies**, built a **simulator** for external testing, and made **code-level optimizations** to support **2500 TPS** at scale.
- Built and own end-to-end **Open Loop Wallet (OLW)** and **Closed Loop Wallet (CLW)** payment flows from the ground up in a **microservices-based architecture**, enabling **INR 10Cr+ transaction volume in the last 3 months** with a **99.6% success rate**.
- Engineered a reusable **AES–RSA hybrid encryption framework** with **Base64 encoding**, securing sensitive payloads **in transit and at rest** across **10+ internal services**; designed for **idempotent, fault-tolerant** inter-service communication.
- Automated secure bank data ingestion pipelines using **Python**, processing **password-protected ZIP files** and **cell-level encrypted Excel/CSV data**, improving data processing efficiency by **60%** and eliminating manual intervention.
- Developed **Grafana-based observability dashboards** across **10+ microservices**, monitoring **API throughput**, **CPU**, and **memory utilization**; proactive alerting reduced incident detection and response time by **40%**.

Associate Software Engineer

Jul 2023 – Apr 2025

- Enhanced **company-wide distributed logging pipeline** using **Fluent Bit**, **Kafka**, **FluentD**, **S3**, and **OpenSearch**, reducing infrastructure and operational costs by **80–90%** while significantly improving production troubleshooting efficiency.
- Built a unified log retrieval system integrating **current logs** with **historical archives** by restoring data from **Amazon S3 Glacier** and indexing into **OpenSearch**, improving search efficiency by **60–70%** and enabling seamless forensic analysis.
- Migrated legacy **cron-based workflows** to **Dolphin Scheduler**, extending **RBAC** with a custom “**Maintainer**” role and integrated **Hub-based authentication** for improved platform security.
- Improved code quality and security posture by increasing **SonarQube coverage to 90%+**, achieving an **A+ security rating** and ensuring long-term maintainability across services.

Software Engineer Intern

Jan 2023 – Jun 2023

- Modernized the company-wide logging pipeline by migrating from **Logstash** to a **Fluent Bit / FluentD + Kafka**-based architecture, significantly reducing resource usage and improving log ingestion performance.

Technical Skills

Languages: Java, Python, PHP, C++, SQL

Backend & Messaging: Kafka, Redis, REST APIs, Event-Driven Architecture

Distributed Systems: Microservices, Idempotency, Fault Tolerance, Distributed Transactions

Cloud & Infra: AWS (S3, EC2, EventBridge, S3 Glacier), Localstack

Observability: Fluent Bit, FluentD, Logstash, OpenSearch, Grafana

Testing & Quality: JUnit, Mockito, SonarQube

AI Tools: Cursor, Claude, GPT Codex, Antigravity

Education

Jadavpur University

2019 – 2023

B.E. in Electronics and Instrumentation Engineering (CGPA: 8+)

Kolkata, India

Barasat Mahatma Gandhi Memorial High School

2011 – 2019

(Class X: 87.4% — Class XII: 80%)

Kolkata, India

Achievements

ThankU Award (CEO), PayU — Recognised by the CEO for delivering **80–90% infrastructure cost reduction** through a company-wide logging pipeline overhaul. [**Certificate**]